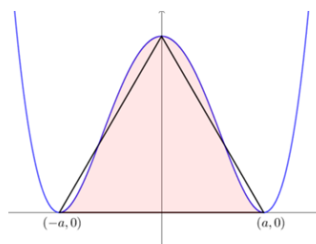


Question 9 – Stage 1



A quartic function has two minimum turning points on the x -axis as shown and the three turning points are at the corners of an equilateral triangle.

If the area of the region under the curve as shown is exactly 1 square unit more than the area of the equilateral triangle, what is the value of a to 3d.p.?

Working

$$\begin{aligned}
 \int_{-a}^a b(x+a)^2(x-a)^2 dx &= a\sqrt{(2a)^2 - a^2} + 1 \\
 2 \int_0^a b(x+a)^2(x-a)^2 dx &= a^2\sqrt{3} + 1 \\
 2 \int_0^a b(x^4 - 2a^2x^2 + a^4) dx &= a^2\sqrt{3} + 1 \\
 2b \int_0^a (x^4 - 2a^2x^2 + a^4) dx &= a^2\sqrt{3} + 1 \\
 2b \left[\frac{1}{5}x^5 - \frac{2}{3}a^2x^3 + a^4x \right]_0^a &= a^2\sqrt{3} + 1 \\
 2b \left(\frac{1}{5}a^5 - \frac{2}{3}a^5 + a^5 \right) &= a^2\sqrt{3} + 1 & ba^4 = a\sqrt{3} \\
 \frac{16}{15}ba^5 &= a^2\sqrt{3} + 1 & b = a^{-3}\sqrt{3} \\
 \frac{16\sqrt{3}}{15}a^2 &= a^2\sqrt{3} + 1 \\
 \frac{\sqrt{3}}{15}a^2 &= 1 \\
 a^2 &= 5\sqrt{3} \\
 a &= 2.94283\dots \\
 a &\approx 2.943
 \end{aligned}$$